
EDCO: Dynamic Curriculum Orchestration for Domain-specific Large Language Model Fine-tuning

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Abstract

Domain-specific large language models (LLMs), typically developed by fine-tuning a pre-trained general-purpose LLM on specialized datasets, represent a significant advancement in applied AI. A common strategy in LLM fine-tuning is curriculum learning, which pre-orders training samples based on metrics like difficulty to improve learning efficiency compared to a random sampling strategy. However, most existing methods for LLM fine-tuning rely on a static curriculum, designed prior to training, which lacks adaptability to the model’s evolving needs during fine-tuning. To address this, we propose EDCO, a novel framework based on two key concepts: *inference entropy* and *dynamic curriculum orchestration*. Inspired by recent findings that maintaining high answer entropy benefits long-term reasoning gains, EDCO prioritizes samples with high inference entropy in a continuously adapted curriculum. EDCO integrates three core components: an efficient entropy estimator that uses prefix tokens to approximate full-sequence entropy, an entropy-based curriculum generator that selects data points with the highest inference entropy, and an LLM trainer that optimizes the model on the selected curriculum. Comprehensive experiments in communication, medicine and law domains, EDCO outperforms traditional curriculum strategies for fine-tuning Qwen3-4B and Llama3.2-3B models under supervised and reinforcement learning settings. Furthermore, the proposed efficient entropy estimation reduces computational time by 83.5% while maintaining high accuracy. The code is available at <https://github.com/GTS-AIData/EDCO>.

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1. Introduction

Enabling large language models (LLMs) to handle tasks of diverse domains represents a hallmark of machine intelligence (OpenAI, 2023; Google, 2023). Research has recently shifted toward developing domain-specific LLMs, yielding notable applications in fields such as medicine, law, and communication (Wang et al., 2025b; Shu et al., 2024; Zhang et al., 2025b). A common approach to constructing such models is fine-tuning a general-purpose pre-trained LLM on specialized datasets. While conventional fine-tuning typically employs random data sampling, emerging evidence indicates that the fine-tuning efficacy is constrained by the training curriculum (Chen et al., 2025), i.e., the order in which training samples are presented. *This is particularly important for fine-tuning domain LLMs because high-quality domain data is typically scarce and costly.*

However, most existing curriculum learning (CL) strategies for LLMs rely on static data ordering, determined based on heuristic metrics such as difficulty or perplexity (Kim & Lee, 2024). Such fixed curricula remain unchanged throughout training, failing to adapt to the model’s evolving ability and knowledge acquisition dynamics, limiting potential gains in both efficiency and final performance. A notable exception is SEC (Chen et al., 2025), which introduces a learnable curriculum policy for curriculum selection. However, its effectiveness is hindered by training instability, as it requires optimizing an additional curriculum policy formulated as a bandit problem.

To address these limitations, we propose Entropy-based Dynamic Curriculum Orchestration (EDCO) method, which continuously adapts the training curriculum to the model’s evolving learning status. EDCO is grounded in two key ideas: *inference entropy* as a measure of sample impact, and *dynamic curriculum orchestration*. Inspired by recent findings that the model’s quick entropy collapse leads to performance saturation (Cui et al., 2025), EDCO prioritizes samples that maximize inference entropy throughout the training process. This ensures that the model is consistently exposed to data points that challenge its current capabilities and reduce uncertainty most effectively. The EDCO framework integrates three core technical components: (1) *an*

efficient entropy estimation module. Due to the computational costs for sweeping over the whole dataset, EDCO uses only prefix tokens to approximate the full-sequence entropy, clearly reducing computational overhead; (2) a *dynamic curriculum generator* that constructs training batches by selecting instances with the highest estimated inference entropy at each training stage; and (3) an *LLM fine-tuning* model for optimizing the LLM. We evaluate EDCO extensively across communication, medical, legal, mathematical reasoning, and summarization tasks. The experiment results demonstrate that EDCO is compatible with supervised and reinforcement learning manners, consistently improving the performance of various models in domain-specific fine-tuning and broader generation settings.

The contributions of this work are summarized as follows. We leverage the key insight that *entropy collapse* hinders model learning to propose EDCO, a dynamic curriculum framework. By actively orchestrating training samples to maintain high inference entropy, EDCO prevents premature convergence and sustains effective exploration throughout the fine-tuning process. Besides, we propose prioritizing high inference entropy samples in a *reverse curriculum pattern*, departing from traditional *easy-to-hard* curricula (Kim & Lee, 2024), and introduce a novel efficient entropy estimation technique that reduces computational overhead while preserving accuracy. Moreover, we demonstrate extensive validation and broad applicability through experiments across diverse domains and task formats, including domain QA, mathematical reasoning, and summarization, showing consistent performance gains under supervised and reinforcement learning-based fine-tuning paradigms.

2. Background

2.1. Problem Formulation and LLM fine-tuning

Consider we have a domain-specific dataset $\mathcal{D} = \{(x, y)_i\}_{i=1}^M$ and a pre-trained LLM $\mathcal{M}_\theta$ parameterized by θ . Here, x is the input prompt (typically a question), and y is the target answer. For simplicity, we use $y \sim \mathcal{M}(\cdot|x)$ to denote sampling an answer y from $\mathcal{M}$ given the question x . The primary objective is to optimize the LLM to achieve high answer accuracy on an unseen dataset, represented as $\mathcal{D}'$. LLMs are typically pre-trained on large-scale corpora to acquire general linguistic capabilities. To adapt them to specific domains or tasks, a common approach is to perform continual pre-training followed by fine-tuning on domain-specific datasets. Two primary fine-tuning paradigms are supervised fine-tuning (SFT) and reinforcement learning fine-tuning (RLFT).

In SFT, the pre-training model is further trained on a curated dataset of input-output pairs specific to a target domain. The training objective is to minimize the cross-entropy loss

between the model’s predictions and the target answers:

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{SFT}}} \left[\sum_{t=1}^T \log \mathcal{M}_\theta(y_t|x, y_{<t}) \right], \quad (1)$$

where x is the input prompt, y is the target sequence, T denotes the sequence length, y_t denotes the t -th token, and $\mathcal{M}_\theta$ is the LLM policy parameterized by θ .

While SFT is effective for instruction following (Pang et al., 2023; Wang et al., 2025a) and style adaptation, it relies heavily on the quality and diversity of the labeled data (Pang et al., 2024b; 2025b). In contrast, RLFT leverages RL to optimize the model toward a reward signal (Kou et al., 2026), which can be more flexible and scalable (Pang et al., 2025c), enabling the model to explore diverse solutions. The objective in RLFT is to maximize the expected cumulative reward:

$$J_{\text{RL}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \mathcal{M}_\theta(\cdot|x)} [r(y)], \quad (2)$$

where $r(y)$ is a reward function that evaluates the quality of the generated sequence y , which can be obtained in the form of self-evaluation (Pang et al., 2024a), rule-based (Mu et al., 2024), or verifiable (Su et al., 2025) reward. Common RL algorithms for LLMs include policy gradient (Sutton & Barto, 1998), PPO (Schulman et al., 2017), and group-based variants like GRPO (Shao et al., 2024).

2.2. The Role of Entropy in Maintaining Exploration

A key challenge in RLFT is the fast collapse of the model’s inference entropy, often occurring within the first few hundred training steps (Cui et al., 2025). This leads to overconfident models that fail to explore alternative reasoning paths, hindering the potential policy improvement. Empirical studies demonstrate a strong correlation between entropy collapse and the score, modeled by the relationship:

$$R = -a \exp(\mathcal{H}) + b, \quad (3)$$

where R is the validation performance and $\mathcal{H}$ is the LLM’s inference entropy. This suggests that performance improvements are effectively *traded* for a reduction in entropy, with the performance upper bound approached as entropy is used up ($\mathcal{H} \rightarrow 0$). Under this circumstance, actively maintaining inference entropy is important for sustained exploration and improved generalization. A direct yet promising strategy is prioritizing high-entropy samples during training, ensuring the model encounters challenging examples that hinder entropy collapse and promote ongoing learning.

3. Method

This section will present our primary contribution, EDCO, a novel training framework that dynamically adapts the

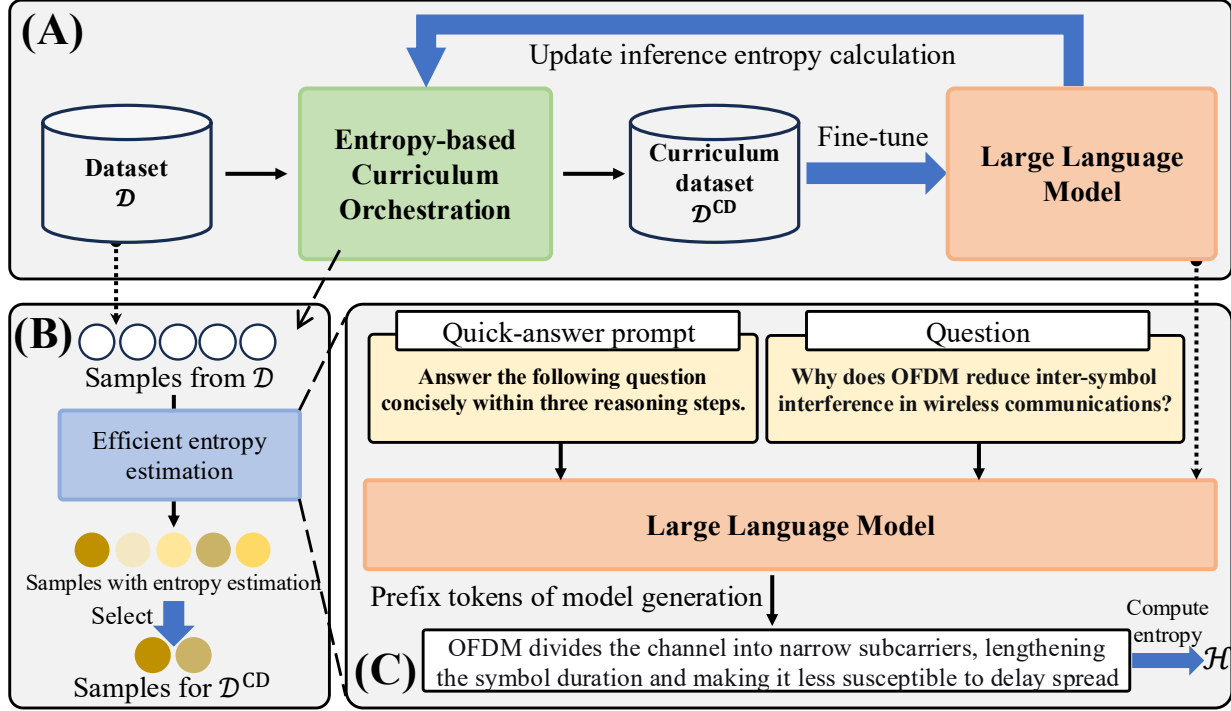


Figure 1. An overview of EDCO method, with three panels: (A) Overall training procedure; (B) Entropy-based curriculum orchestration module that periodically updates the training curriculum; (C) Efficient entropy estimation module that calculates the sample entropy.

curriculum to the model’s evolving learning state through periodic inference entropy estimation. Unlike static curricula that follow a predetermined order, EDCO continuously re-evaluates the training sample’s significance by calculating the inference entropy on samples with current model. As the overall framework of EDCO shown in Fig. 1, the core procedure operates through: (1) computing inference entropy for the remaining high-quality training samples using our efficient estimation techniques; (2) selecting the top- N highest-entropy samples from this pool to form the training curriculum for the next phase; (3) fine-tuning the model on this selected subset; and (4) repeating the process (1-3) until convergence. This dynamic approach is compatible with both SFT and RLFT paradigms, addressing the need for sustained exploration, particularly in RL training where entropy collapse can rapidly hinder progress.

3.1. Dynamic Curriculum Orchestration via Inference Entropy

An important motivation behind EDCO is that maintaining high inference entropy are beneficial to training (Cui et al., 2025), as they represent points of maximum uncertainty encourage the model to explore the solution, rather than exploiting the existing knowledge. This constitutes a reverse curriculum strategy that prioritizes more challenging examples rather than following the conventional easy-to-hard progression. EDCO is dynamically adaptive: as the model

learns and its uncertainty distribution shifts, the curriculum is updated to reflect new challenging frontiers.

Formally, at training interval k , we compute the inference entropy $H(y|x; \theta_k)$ for each sample (x, y) in the training dataset $\mathcal{D}$, where θ_k denotes the model parameters at interval k . The entropy for a given sample is defined as:

$$H(y|x; \theta_k) = -\mathbb{E}_{y \sim \mathcal{M}_{\theta_k}(\cdot|x)} [\log \mathcal{M}_{\theta_k}(y|x)]. \quad (4)$$

Samples are then ranked by their entropy values, and the top N with the highest entropy are selected to form the curriculum dataset $\mathcal{D}_k^{\text{CD}}$ for the next interval:

$$\mathcal{D}_k^{\text{CD}} = \{(x, y) \in \mathcal{D} \mid H(y|p_{\text{quick}}, x; \pi_{\theta_k}) \text{ ranks in top } N\}. \quad (5)$$

The model is subsequently fine-tuned on $\mathcal{D}_k$ for a fixed number of steps or until the next curriculum update. At the first curriculum iteration, entropy is computed using the pretrained or continually pretrained initialization; subsequent updates then track the model’s changing uncertainty profile rather than freezing this initial signal. This periodic reassessment and selection mechanism ensures the training curriculum remains aligned with the model’s continuously evolving capabilities, preventing plateaus and maintaining high learning efficiency throughout the training process.

3.2. Efficient Entropy Estimation with Prefix Tokens

A significant challenge in implementing dynamic curriculum is the computational expense of calculating full-sequence inference entropy across the entire dataset. To address this, we introduce two innovative techniques that substantially reduce computational overhead while preserving estimation accuracy.

3.2.1. QUICK-ANSWER PROMPTING

Traditional LLM inference often involves lengthy chain-of-thought reasoning, which is computationally expensive for entropy estimation. We propose *Quick-Answer Prompting* (QAP) technique to modify the input prompt to encourage the model to output the final answer directly without much reasoning steps. Specifically, instead of using a standard instruction like ‘‘Solve the following problem step by step’’, we use a QAP p_{quick} : ‘‘Answer the following question concisely within three reasoning steps’’. By pushing thinking trace towards the answer, the prefix tokens are more effective in reflecting the model’s understanding of the samples, providing a more efficient and concentrated entropy signal. We investigate the effect of QAP in Appendix C.4.

3.2.2. PREFIX ENTROPY APPROXIMATION

Calculating the exact entropy over the entire output sequence y requires autoregressively generating all tokens and remains prohibitively expensive. We propose *Prefix Entropy Approximation*, which estimates the full-sequence entropy using only the first few tokens of the output. This approach is motivated by the observation that the entropy of the initial tokens strongly correlates with the uncertainty of the complete generation. Specifically, we approximate the full-sequence entropy as:

$$H(y|y_{<t}, p_{\text{quick}}, x; \theta_k) \approx - \sum_{t=1}^L \log \mathcal{M}_{\theta_k}(y_t|y_{<t}, p_{\text{quick}}, x), \quad (6)$$

where L is a small fixed number of prefix tokens (e.g., $L = 50$). This approximation reduces the computational complexity from $O(T)$ to $O(L)$ per sample, where T is the average output length. Our experiments demonstrate that this prefix-based entropy maintains a strong rank correlation with the full-sequence entropy, ensuring reliable sample selection while achieving significant speedups.

3.3. Compatibility with Fine-Tuning Paradigms

EDCO is agnostic to the underlying fine-tuning algorithm, which can be naturally integrated with SFT and RLFT frameworks. In SFT, the training objective on the selected high-entropy subset $\mathcal{D}_k$ remains the standard cross-entropy loss in Eq. (1). The dynamic curriculum ensures that the model focuses on samples that are currently most challenging, pre-

Algorithm 1 EDCO

Require: The preprocessed high-quality dataset $\mathcal{D}$, initial model parameters θ_0 , prefix length L .

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1:  $k \leftarrow 0$ 
2: while not converged do
3:    $\mathcal{D}_k \leftarrow \emptyset$ 
4:   for each sample  $(x, y) \in \mathcal{D}$  do
5:     Construct quick-answer prompt  $x' \leftarrow (p_{\text{quick}}, x)$ 
6:     Compute prefix entropy  $H \leftarrow - \sum_{t=1}^L \log \mathcal{M}_{\theta_k}(y_t|y_{<t}, x')$ 
7:     Add  $(x, y, H)$  to  $\mathcal{D}_k$ 
8:   end for
9:   Sort  $\mathcal{D}_k$  by entropy in descending order
10:  Select top  $N$  samples as the curriculum dataset  $\mathcal{D}_k^{\text{CD}}$ 
11:  Fine-tune  $\theta_k$  on  $\mathcal{D}_k^{\text{CD}}$  for  $N$  steps using SFT (Eq. (1)) or RLFT (Eq. (2))
12:   $k \leftarrow k + 1$ 
13: end while
14: Return  $\theta_k$ 
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venting overfitting to easy patterns and promoting broader generalization. In RLFT, EDCO addresses the critical issue of entropy collapse by continuously supplying high-entropy samples that encourage exploration. The RL objective (Eq. (2)) is applied to the selected subset. By maintaining exposed to high inference entropy examples, EDCO effectively delays entropy collapse, allowing the model to explore a wider range of behaviors and discover superior policies.

3.4. Algorithm Summary

Algorithm 1 summarizes the complete EDCO training procedure. The process begins with the preprocessed dataset $\mathcal{D}$ and initial model parameters. At each curriculum update interval, we compute the inference entropy for all samples using quick-answer prompting and prefix entropy approximation. The top N highest-entropy samples are selected to form the training batch for the subsequent phase. The model is then fine-tuned on this subset with either SFT or RL objectives. This cycle repeats until training convergence, ensuring the model is consistently challenged by appropriately difficult examples throughout the learning process.

4. Related Work

Our work lies at the intersection of domain-specific adaptation for large language models and curriculum learning. Accordingly, we review relevant literature in four areas.

4.1. Domain-specific Large Language Models

Recent advances have shown the growing attention on domain-specific LLMs across various professional fields

where accuracy and specialized reasoning are critical requirements (Song et al., 2025; Jeong, 2024; Pal et al., 2024). In medicine, LLMs have evolved from basic information retrieval tools to sophisticated clinical reasoning systems capable of supporting complex diagnostic processes (Berger et al., 2025). Similarly, researchers in the legal domain have explored numerous LLMs applications for document analysis, case prediction, and legal reasoning. However, challenges remain in handling complex domain-specific relationships that general models often misunderstand (Colombo et al., 2024). The adaptation of general-domain LLMs for specialized applications in law typically focuses on fine-tuning approaches rather than introducing new architectural innovations (Chen et al., 2024). In communication systems, recent surveys have investigated the integration of LLMs across different network domains, including mobile networks and related technologies, highlighting both opportunities and challenges in this emerging field (Boateng et al., 2024). However, these approaches ignore the training curriculum structure, treating all samples equally valuable regardless of the model’s evolving proficiency.

4.2. Uncertainty-Driven and Entropy-Based Data Selection

Uncertainty quantification has become a pivotal metric for evaluating LLM reliability and data quality. Recent works have leveraged entropy and confidence scores for various selection tasks. For instance, Liang et al. (2025) utilize predictive entropy to identify unreliable responses in medical VLMs. Similarly, Zhang et al. (2025a) introduce Long-text Uncertainty Quantification to enhance selective question answering. However, these approaches typically apply uncertainty metrics either as a static pre-filtering step (Liu et al., 2025) or for inference-time control (Agrawal et al., 2025), rather than as a dynamic signal to guide the training trajectory. Unlike active learning methods that focus on selecting unlabeled data for annotation to reduce labeling costs (Pang et al., 2025a), EDCO focuses on training efficiency by dynamically re-weighting existing data based on the model’s real-time inference entropy, ensuring the model always learns from samples at its capability frontier.

4.3. Curriculum Learning for Large Language Models

CL has emerged as a promising approach to improve the efficiency and effectiveness of LLM training. Traditional curriculum strategies often follow an “easy-to-hard” progression, starting with simpler tasks and gradually introducing more complex examples (Kim & Lee, 2024). Some approaches have utilized data distribution characteristics to determine sample ordering, with Static DDCL (Chaudhry & Sharma, 2025) representing an innovative method in utilizing data distribution for curriculum organization. More recently, researchers have begun exploring dynamic cur-

riculum approaches that adapt during training, such as the framework combining CL with LLM reasoning that allows for adaptive adjustment of difficulty levels based on model performance (Zhang et al., 2024b). To overcome the limitations of static curricula, dynamic data selection strategies have gained attention. DEITA (Liu et al., 2024) studies automatic data selection for instruction tuning, while DOTS (Sun et al., 2025) targets data efficiency in reinforcement fine-tuning through adaptive difficulty prediction and replay. Hübötter et al. (2024) propose active fine-tuning for test-time adaptation, while Middo (Tang et al., 2025b) introduces a model-informed data optimization loop to enhance fine-tuning quality. Reverse curriculum approaches (Florensa et al., 2017) in reinforcement learning have also demonstrated potential by starting with more complex examples and progressing backward. Yet, these methods remain unexplored for domain-specific LLM fine-tuning. Crucially, no prior work dynamically reorders samples based on the model’s instantaneous uncertainty during fine-tuning, an essential factor for maximizing learning efficiency in data-constrained domains.

4.4. Entropy as a Learning Signal

Entropy has gained attention as a valuable signal for guiding LLM optimization (Tang et al., 2025a). Microscopic Strategy on Responses method (Li et al., 2025) has shown that high entropy in token selection corresponds to greater diversity in training samples, which can make LLM training more robust and less prone to overfitting. Several studies have explored entropy-based data selection techniques to effectively reduce the amount of training data required while maintaining performance (Yin et al., 2024). In RL contexts for LLMs, entropy-based terms have served as robust, self-regularization signals that guide learning without altering the original gradient flow of the base model (Cheng et al., 2025). The EDT method (Zhang et al., 2024a) has also investigated dynamic adjustment of LLM decoding behavior based on confidence metrics related to entropy. Vocabulary curriculum methods (Yu, 2025) have employed entropy-guided expansion strategies to enable models to learn transferable representations more effectively. Despite these advances, the application of inference entropy as a dynamic curriculum generation mechanism for domain-specific LLM fine-tuning remains largely underexplored, particularly in contexts where SFT and RL training are required.

5. Experiment

In this section, we conduct extensive experiments to verify the effectiveness of EDCO for domain-specific LLM fine-tuning. We conduct experiments across diverse challenging domains, *data communication*, *wireless communication*, *legal* and *medicine* domains, to answer the following key re-

search questions: (1) How does EDCO perform compared to existing curriculum learning methods for LLM fine-tuning (Sec. 5.2)? (2) Does EDCO generalize beyond domain QA to reasoning and summarization tasks (Sec. 5.2)? (3) What is the underlying mechanism of dynamic curriculum orchestration (Sec. 5.3)? (4) How effective and efficient is the proposed entropy estimation module (Sec. 5.4)?

5.1. Experimental Setting

Datasets and domains. We evaluate EDCO mainly on challenging communication domains: *Data Communication* (Datacom) and *Wireless Communication* (Wireless). We construct a specialized dataset for each domain comprising 12,000 question-answer pairs, synthesized from a diverse corpus of product documentation, technical solutions, and domain knowledge bases. The datasets encompass diverse question types, including single-choice, multiple-choice, and open-ended QA, covering fundamental principles, product concepts, terminology understanding, and multi-step reasoning tasks. These training dataset could be utilized for SFT and RLFT. All methods are evaluated on a held-out test set of 230 challenging, unseen problems from the same domains. In addition to communication domains, we involve datasets from medicine (MedQA (Jin et al., 2021)) and legal (JEC-QA (Zhong et al., 2020)) domains to provide more comprehensive evaluation. We further evaluate on Math-17k (Yu et al., 2025) and CNN/DailyMail summarization (Nallapati et al., 2016) to test whether the curriculum principle extends beyond domain QA. We provide more details about the dataset and domain description in Appendix B.1.

Baseline for comparison. We compare EDCO against representative baselines with both static and dynamic curriculum learning strategies: (1) **Random Sampling (RS)**: The standard, curriculum-free approach for LLM fine-tuning; (2) **Length-based Curriculum (Length)**: A simple heuristic ordering samples by input sequence length (easy-to-hard based on brevity). (3) **Answer Complexity (AC)**: A heuristic that orders samples by the number of sentences in the answer, representing reasoning depth. (4) **Perplexity-based Curriculum (PPL)**: A classic model-based approach representing the "easy-to-hard" paradigm, where difficulty is determined by a pre-trained model’s perplexity (Hu et al., 2024). (5) **Self-evolving curriculum (SEC)** (Chen et al., 2025): Learn a curriculum policy with UCB algorithm (Auer, 2002) to select the training batch. (6) **Dynamic-PPL**: A matched-condition dynamic version of PPL, which uses the same curriculum update interval as EDCO but replaces entropy with perplexity. We also compare with stronger recent data-selection methods, including DEITA (Liu et al., 2024) and DOTS (Sun et al., 2025), on public-domain experiments.

Implementation details. All experiments are implemented

using the MindSpeed-RL framework (Feng et al., 2025). We use Qwen3-4B and Llama3.2-3B as the base model to demonstrate applicability across model scales and types. For RLFT, we employ the GRPO algorithm (Shao et al., 2024) to train the language models. The rewards are generated by Deepseek-V3 (DeepSeek-AI, 2024) performing automated verification of model responses against ground-truth answers for wireless and datacom domains, and generated from rule-based verification for medical and legal domains. All experiments are conducted on a computing cluster with 256 KUNPENG 920 CPU cores and 8 Ascend 910B3 NPUs. Appendix B.4 lists the hyper-parameters used for experiments.

5.2. Main Results for RLFT and SFT

Results for reinforcement learning fine-tuning. Fig. 2 (A) and (B) present the RLFT performance with the Qwen3-4B model. Before the RLFT, we perform continued pre-training with the data of communication domain for all methods. In the Datacom domain, while all CL methods manage to improve upon the base model, EDCO achieves a higher accuracy of 46.96%, outperforming both RS (40.43%) and the PPL (44.78%). Across three independent evaluations, EDCO also obtains the highest mean score with lower variance than PPL, as reported in Appendix C.1. This indicates that dynamic curriculum orchestration effectively accelerates policy optimization in complex environments. Conversely, the Wireless domain exposes the instability of standard RLFT. Static strategies like RS and AC fail to yield positive training, resulting in performance degradation compared to the base model (35.22% vs. 34.35%/33.91%). In contrast, EDCO demonstrates superior performance, consistently securing the highest score of 38.70% across both domains. This result supports our claim that relying on static metrics or RS is insufficient for RLFT, whereas maintaining high inference entropy through dynamic scheduling provides the necessary stable learning signals.

Results for supervised fine-tuning. We observe a similar trend in the SFT experiments, verifying that the benefits of EDCO extend across different training paradigms. As shown in Fig. 2(C) and (D), EDCO consistently achieves the highest accuracy, reaching 33.7% on Wireless and 36.3% on Datacom. Notably, it outperforms the PPL-based curriculum by margins of 2.0% and 3.3% respectively, showcasing a clear advantage over static model-based approaches. Besides, several *easy-to-hard* baselines (Length, AC, PPL) fail to improve upon or even degrade performance compared to the base model on the Wireless dataset. This reveals a critical pitfall: a poorly designed static curriculum can be actively detrimental to learning, especially in specialized domains where syntactic simplicity does not equate to conceptual ease. For instance, the poor performance of the AC method in Datacom SFT (28.3%) highlights how a static

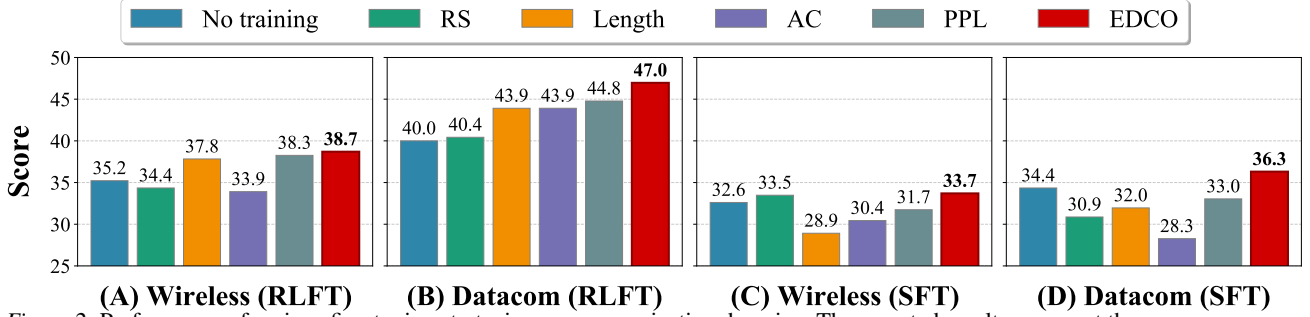


Figure 2. Performance of various fine-tuning strategies on communication domains. The reported results represent the answer accuracy, averaged over three evaluations.

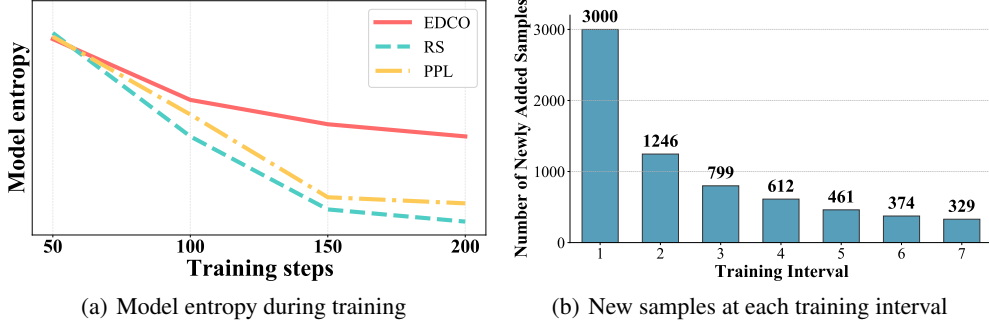


Figure 3. Analysis of the training process of EDCO method. (A) The model’s inference entropy during the training. (B) The number of first-time samples added in each training interval.

focus on syntactically complex answers fails to adapt to the model’s growing knowledge, leading it to struggle with samples that remain too difficult repeatedly. These results prove that dynamic curriculum orchestration is a more robust and effective strategy across different training paradigms.

Table 1. Performance comparison on medical and legal domains using Llama3.2-3B.

Dataset	No Training	RS	Length	AC	PPL	EDCO
MedQA	32.1	32.9	35.1	32.4	24.6	36.7
JEC-QA	16.2	16.2	10.5	14.6	12.4	17.4

RLFT on medical and legal domains. To demonstrate the generalizability of EDCO beyond telecommunications, we extended our evaluation to two qualitatively different domains using the Llama3.2-3B model (Grattafiori et al., 2024). As the result shown in Tab. 1, EDCO consistently outperforms RS and static curricula (Length, AC, PPL) in these new domains. Notably, on MedQA, EDCO achieves 36.7% accuracy compared to 32.9% for RS and 24.6% for PPL. Similarly, on JEC-QA, EDCO leads with 17.4%. These results validate that prioritizing high inference entropy is a fundamental principle for efficient fine-tuning across diverse fields and is effective across different model architectures (Qwen and Llama) and sizes (3B and 4B).

Comparison with stronger public baselines. To address whether the improvement holds beyond traditional static

heuristics, we compare EDCO with DEITA (Liu et al., 2024) and DOTS (Sun et al., 2025) on MedQA. DEITA is a stronger static data-selection baseline, while DOTS represents a recent dynamic difficulty-targeted method. As shown in Tab. 2, EDCO achieves the best accuracy, indicating that entropy-based dynamic orchestration remains effective against stronger recent selection strategies.

Table 2. Comparison with stronger static and dynamic baselines on MedQA.

Dataset	No Training	RS	DEITA	DOTS	EDCO
MedQA	32.1	32.9	34.9	35.0	36.7

Generalization to reasoning and summarization. We further evaluate EDCO on Math-17k and CNN/DailyMail to test whether the high-entropy curriculum principle extends beyond domain-specific QA. As shown in Tab. 3, EDCO outperforms random sampling on both mathematical reasoning and open-ended summarization. These results broaden the empirical evidence, while we do not claim exhaustive coverage of all LLM task families.

Comparison with dynamic baselines. To further verify EDCO’s effectiveness, we compared it against two dynamic curriculum methods on the Datacom domain using the Qwen3-4B model: SEC and Dynamic-PPL. Dynamic-PPL serves as a matched-condition ablation: it keeps the

Table 3. Generalization beyond domain QA with Qwen3-4B.

Task	Metric	No Training	RS	EDCO
Math-17k	Accuracy	0.673	0.713	0.723
CNN/DailyMail	ROUGE	0.29	0.40	0.42

same dynamic curriculum update mechanism and interval as EDCO, and only swaps the selection signal from inference entropy to perplexity. As shown in Tab. 4, EDCO clearly outperforms Dynamic-PPL (47.0% vs. 41.3%), indicating that the gain is mainly from the entropy-based selection signal rather than dynamic updating alone. While perplexity is often dominated by surface token prediction statistics, inference entropy better captures the model’s current uncertainty over generated behavior. Furthermore, EDCO outperforms the bandit-based SEC method (34.78%), which suffered from instability during the policy learning phase in this setting.

Table 4. Comparison against learnable and dynamic baselines on the Datacom domain (Qwen3-4B). EDCO outperforms both the bandit-based SEC and the dynamic perplexity strategy.

Domain	No Training	RS	SEC	Dynamic-PPL	EDCO
Datacom	40.0	40.4	34.78	41.3	47.0

5.3. Analysis of the Training Process

The previous subsection presents that EDCO achieves better performance for domain LLM fine-tuning. Now we investigate the mechanisms behind EDCO’s superior performance through detailed training process analysis.

Entropy change during the training process. The motivation behind EDCO is to maintain high inference entropy during the training. Fig. 3(a) validates this principle empirically. Specifically, during the training process, we record the model’s inference entropy. While the model trained with random sampling sees its entropy decay rapidly, EDCO successfully sustains a high-entropy, high-challenge learning environment throughout training. This demonstrates that our dynamic selection process prevents the model from settling into a low-uncertainty state, constantly pushing it to refine its understanding of more complex or nuanced samples. This sustained challenge directly correlates with its superior final performance.

Curriculum selection dynamics. Fig. 3(b) visualizes the composition of the training curriculum at each update interval. The numbers in the figure stand for the number of samples that have never been selected for training previously. Appendix C.2 further shows that high-entropy samples consistently have lower model accuracy than low-entropy samples, supporting entropy as a useful signal for current sam-

ple difficulty. The analysis reveals that the curriculum is *constantly evolving*. At each interval, EDCO strategically selects a mix of entirely new, high-entropy samples alongside previously seen samples that remain challenging (i.e., still exhibit high entropy) for the model’s current state. This dynamic ensures that complex concepts are not prematurely discarded but are revisited until mastered, while simultaneously introducing new challenges to broaden the model’s knowledge. This adaptive re-challenge mechanism is a key difference from static curricula, which follow a rigid, one-pass sequence.

5.4. Effectiveness of Entropy Estimation & Ablation Study

We verify the effectiveness of the entropy estimation in EDCO through two dimensions: *accuracy* and *efficiency*.

Accuracy of Prefix-based Estimation. Fig. 4(a) compares the entropy calculated using only a 128-token prefix against the entropy from the full sequence. Since EDCO uses entropy to rank samples for curriculum selection, rank correlation is the most relevant reliability measure. The results show a strong positive correlation, with a *Spearman coefficient* of 0.7333 ($p = 0.0158$). This result confirms that the prefix-based approximation preserves the ordering needed for reliable sample selection, validating EDCO to reduce computational cost without decreasing the integrity of the curriculum signal.

Computational efficiency and wall-clock time. As detailed in Tab. 5, the prefix-based estimation leads to a clear efficiency gains. It reduces the per-sample estimation time from 2.24s to just 0.37s, an 83.5% reduction in computational overhead. This speedup transforms dynamic curriculum generation from a computationally prohibitive method into a practical and scalable strategy. Furthermore, when parallelized across 8 NPUs, the estimation time reduces to 0.04 seconds per sample, making near-real-time curriculum updates feasible even for large datasets. Besides, we record that the wall-clock time for EDCO’s training for 500 RL steps is 33.58 hours, compared to RS’s 30.67 hours (~10% difference). The overhead of entropy estimation is generally covered by the RL training overhead. Thus, the entropy estimation introduce acceptable overhead for the training.

Table 5. Computational efficiency of entropy estimation methods (seconds per sample).

Method	Single process	Parallel with 8 cards
Full-sequence	2.24	0.24
Prefix-based (Ours)	0.37	0.04

Effect of the number of prefix tokens. We conduct an ablation study on prefix token length to provide practical implementation guidelines, with the entropy estimation results shown in Fig. 4(b). The analysis shows that while

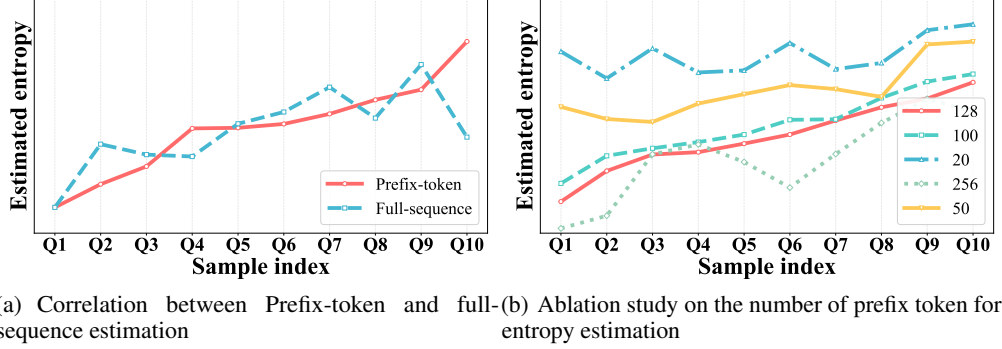


Figure 4. Analysis of the efficient entropy estimation module in EDCO. (A) Comparison of entropy estimation using a 128-token prefix versus the full sequence. (B) Ablation study on the prefix length. To better visualize data trends, the sample indices are arranged in ascending order of the entropy estimated with the 128-token prefix. Ablation study on QAP is in Appendix C.4.

a very short prefix (e.g., 20 tokens) can lead to unstable estimations, the entropy trends stabilize significantly for prefixes of 50 tokens or more. This indicates that a prefix length of 50-128 tokens strikes an optimal balance between estimation stability and computational efficiency, offering a robust default configuration for future applications.

6. Conclusion

This work addresses the challenge of efficiently specializing LLMs for specific domains, where data scarcity demands maximally effective fine-tuning strategies. We propose EDCO, a novel framework that introduces a dynamic, entropy-driven curriculum to continuously align the training process with the model’s evolving learning state. The key contribution lies in shifting away from static curricula by prioritizing samples with maximum inference entropy, which is efficiently implemented via quick-answer prompting and prefix-based entropy approximation. Extensive experiments across communication, medicine, law, mathematical reasoning, and summarization tasks demonstrate that EDCO consistently enhances performance under both SFT and RLFT paradigms. However, there are still some limitations. First, the efficiency of entropy estimation, while improved from prefix token approximation, still introduces periodic computational overhead compared to standard fine-tuning. Future work could explore more lightweight techniques to predict sample significance without full forward passes. Second, the current method operates on a fixed update interval for curriculum orchestration. An adaptive scheduling mechanism, triggered by performance threshold or entropy convergence, could further optimize the training dynamics. Finally, although we extend the evaluation beyond domain QA to reasoning and summarization, broader validation on tasks such as code generation, long-context understanding, low-resource languages, and highly technical scientific fields remains an important direction. We believe these interesting directions are worth further exploration for developing more powerful and efficient domain-specific LLMs.

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Impact Statement

This paper presents work whose goal is to advance the field of machine learning by improving the efficiency and adaptability of LLM fine-tuning. A primary societal consequence of our work is the potential for reduced computational resource consumption during model training; specifically, our efficient entropy estimation significantly lowers compute time, contributing to more sustainable and energy-efficient AI development. While our work aims to enhance model performance in high-stakes domains such as medicine and law, the broader ethical implications, such as the risks of bias or reliability in generative models, remain consistent with well-established challenges in the field. We do not feel there are other specific negative societal consequences that must be highlighted here.

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Appendix

A. Discussions about Reverse Curriculum Learning

This work’s prioritization of high-entropy samples represents a departure from the traditional easy-to-hard CL paradigm, effectively creating a reverse curriculum. This section provides additional discussions for this choice in the context of fine-tuning pre-trained LLMs within specific domains. Traditional CL is inspired by human’s education process, where a learner starts with foundational concepts and gradually progresses to more complex topics. This is effective when training a model from scratch, as it stabilizes the initial learning stages and prevents divergence caused by overly challenging samples.

However, fine-tuning a pre-trained LLM presents a different scenario. The goal of fine-tuning is not to teach the model basic concepts but to specialize its existing knowledge for a specific domain. In this context:

- **“Easy” samples offer diminishing returns.** Samples that the model can already answer with low uncertainty (low entropy) are largely redundant with its pre-existing knowledge. Training on them provides a weak learning signal (i.e., small gradients) and does little to refine its domain-specific abilities.
- **“Hard” samples are most informative.** Samples with high inference entropy are precisely those where the model’s general knowledge is insufficient or conflicts with domain-specific nuances. These are the points of highest uncertainty and, therefore, the greatest potential for information gain. By focusing on these samples, EDCO ensures that each training step is maximally efficient at reducing the model’s domain-specific predictive uncertainty.

Thus, for domain specialization, a dynamic curriculum that consistently presents the most challenging material, as measured by the model’s current state, is more effective than a static, easy-to-hard progression. EDCO’s approach ensures the model is perpetually operating at the frontier of its competence, accelerating its adaptation to the target domain.

B. More Experiment Details

B.1. Details about the Dataset for Evaluation.

Wireless. The wireless domain is a key branch of information technology, focusing on the research, development, and application of wireless communication technologies. This domain covers a variety of technologies and products, such as 5G, 4G, Wi-Fi, Bluetooth, and NFC, which together build a modern wireless communication infrastructure that supports a variety of application scenarios from mobile communications and the IoT (Internet of Things) to smart homes. 5G technology provides higher data transmission rates, lower latency, and higher connection density, supporting large-scale IoT applications and HD video transmission. Wi-Fi technology is widely used in homes, offices, and public places to provide high-speed wireless network connections.

Datacom. The data communication domain is a vital branch of information technology, focused on the efficient and secure transmission of data. This domain includes various technologies and products, such as network devices like routers, switches, firewalls, and associated software solutions. Together, these technologies and products form the foundation of modern network infrastructure, supporting a wide array of applications, from home broadband connections to enterprise-level data centers. The field of data communication is constantly evolving, with emerging technologies like software-defined networking and network functions virtualization driving transformations in network architecture, enhancing flexibility and manageability.

In addition, in order to better evaluate the model capability, high-quality testing datasets of two communication domains are constructed. In the first step, the original corpus is collected and obtained through manual quality inspection. Then, we use the LLM to synthesize objective questions such as selection and judgment based on these corpus. Subsequently, the LLM and manual quality inspection are used to review and filter the synthesized objective questions from key dimensions such as question integrity and answer accuracy. Finally, 230 evaluation data samples are constructed for each domain.

MedQA (Jin et al., 2021). It is a large-scale, open-domain question-answering dataset designed to evaluate the professional medical knowledge and clinical reasoning capabilities of AI models. Collected from professional medical board examinations,

the dataset covers three languages: English, Simplified Chinese, and Traditional Chinese. The English version is derived from the United States medical licensing examination. The questions are presented in a multiple-choice format, typically featuring a clinical vignette (a detailed patient case description) followed by a question and four or five answer options. To solve these problems, models must possess extensive domain-specific knowledge and the ability to perform multi-hop reasoning over medical concepts to identify the single correct diagnosis or treatment plan.

JEC-QA (Zhong et al., 2020). JEC-QA is a legal-domain question-answering dataset sourced from the National Judicial Examination of China. It consists of approximately 26,000 multiple-choice questions that require deep understanding of Chinese legal provisions and complex logical reasoning. The questions are categorized into two types: knowledge-driven questions, which focus on the definition and interpretation of legal concepts, and case-analysis questions, which require applying legal principles to specific scenarios. A distinct feature of JEC-QA is its answer format, which includes both single-answer and multiple-answer questions (one or more of the four options), significantly increasing the difficulty for model evaluation.

Math-17k (Yu et al., 2025). Math-17k is a mathematical reasoning dataset used to evaluate whether the model can solve verifiable reasoning problems beyond domain-specific QA. We report answer accuracy after extracting the final answer from the model response.

CNN/DailyMail (Nallapati et al., 2016). CNN/DailyMail is an abstractive summarization benchmark with open-ended generation targets. We use this dataset to evaluate whether entropy-based curriculum orchestration can benefit non-QA generation tasks, and report ROUGE scores for summarization quality.

Evaluation protocol. To develop LLM tailored for the communication field and enhance their performance and applicability in the datacom and wireless domains, we construct SFT datasets for both domains. Leveraging a rich set of original corpora, including product documents and solutions from the datacom and wireless domain, we use the LLM synthesis method to generate 20,000 data samples for each domain. These datasets encompass various formats, including single-choice, multiple-choice, and True/False questions, addressing training requirements such as basic principles, product concepts, terminology understanding, and multi-point knowledge reasoning.

B.2. Prompts Used in Experiments

We provide the prompts used in our experiments as follow.

- **Prompts for quick answer:** Please answer the following question in no more than three sentences. If necessary reasoning is required, please minimize the reasoning process as much as possible. Problems: `\{problems\}`
- **Prompts for Single-choice question:** For the following single-choice question, there is only one correct answer. Please analyze the questions and answer options and place the correct option number in `\boxed{\}`, for example, `\boxed{A}`.
- **Prompts for Multi-choice question:** For the following multiple-choice question, there are only multiple correct answers. Please analyze the questions and answer options and place the correct option numbers in `\boxed{\}`, for example, `\boxed{A, B, D}`.
- **Prompts for True/False question:** For the following question, analyze whether the description in the questions is correct or incorrect, and put the final answer in `\boxed{\}`. If the answer is correct, output `\boxed{correct}`. If the answer is incorrect, output `\boxed{error}`.

B.3. Examples of the Dataset

Tab. B.3 presents the examples from the datasets used for training and evaluation in our experiments.

B.4. Hyperparameters

The hyper-parameters for implementing EDCO and experiments are presented in Tab. 7. When implementing baseline methods, we use the same hyper-parameters as EDCO.

Domain	Example question	Target answer
Wireless-training	“What should I do if the DSP GTP-PATH command output shows that the GTP path is in the DEETECT state when the alarm is generated?”	”...1.First, check whether the peer GSN address specified in the alarm information is valid...;2.Next, execute the PING command to check if the link is normal...;3.Confirm whether the peer GSN can respond to ECHO messages...”
Wireless-testing	“During the deployment and operation of the LMT, the LMT is forcibly connected in HTTPS or WSS mode to ensure secure connection. In this mode, digital certificates are required for authentication. In addition, ... In such a deployment scenario, if the MAE of a carrier is set to HTTP login mode and the LMT is set to forcible HTTPS connection mode, what will happen? A. ...B. If the LMT connection mode is set to Force HTTPS, MAE proxy login fails to access the LMT due to protocol mismatch. The connection cannot be established even if the OM channel is normal. C. ...D. ...”	“The answer is X.”
Datacom-training	“When configuring the Segment VXLAN feature, how can you enable EVPN as the VXLAN control plane on Transit Leaf1 and Transit Leaf2, and configure BGP EVPN peer relationships?”	”...1.Enter the BGP view or BGP multi-instance view...;2.Enter the BGP-EVPN address family view...;3.Configure the split group for BGP EVPN peers (groups)...;4.Enable the function to mark routes received from BGP EVPN peers as re-originated...”
Datacom-testing	“...The device supports creating subinterfaces on Layer 2 Ethernet and Layer 2 Eth-Trunk interfaces for VLAN termination to achieve inter-VLAN forwarding. However, the USG9500 series devices do not support creating subinterfaces on these two types of interfaces.”	“The answer is error”

Table 6. Examples from the datasets used in our experiments.

Table 7. Hyper-parameters for training EDCO and baselines.

Hyper-parameters	Value
Prefix token number	128
Epoch number	2
Batch size	8
SFT learning rate	$1.25e - 6$
SFT learning rate decay style	<i>cosine</i>
Train iteration	3000
Sequence length	4096
RLFT learning rate	$1e - 6$
RLFT learning rate decay Style	<i>constant</i>
RL γ	1
RL λ	0.95
Mini batch size	4
RL clip ratio	0.2

C. Additional Experimental Results

C.1. Statistical Stability Across Runs

To assess whether the improvements are stable across repeated evaluations, we report the standard deviations across three independent runs on the Datacom task in Tab. 8. EDCO achieves the highest mean score while maintaining lower variance than PPL, suggesting that its advantage is not driven by an unstable run.

Table 8. Mean and standard deviation across three independent runs on Datacom.

Method	RS	Length	AC	PPL	EDCO
Final Score	40.4 ± 3.20	43.9 ± 2.54	43.9 ± 1.64	44.8 ± 7.65	47.0 ± 1.89

C.2. Difficulty Analysis of High-Entropy Samples

We directly evaluate whether high-entropy samples are more difficult for the current model. At each curriculum iteration on MedQA with Qwen3-4B, we compare model accuracy on 200 high-entropy and low-entropy samples. As shown in Tab. 9, high-entropy samples consistently yield lower accuracy, indicating that inference entropy tracks the model’s current sample difficulty.

Table 9. Accuracy on high-entropy and low-entropy samples during EDCO training.

Curriculum Iteration	Acc. on high entropy	Acc. on low entropy
1	0.755	0.822
2	0.705	0.785
3	0.735	0.770
4	0.785	0.805
5	0.795	0.815

C.3. Hyperparameter Sensitivity

We evaluate the sensitivity of EDCO to the curriculum size, number of curriculum updates, and prefix length. The results in Tabs. 10–12 show that EDCO remains effective across a range of settings, with larger curriculum subsets, more updates, and longer prefixes generally improving performance.

Table 10. Sensitivity to the number of selected samples in each curriculum.

Samples Size N	200	400	600
Final score	0.801	0.813	0.817

Table 11. Sensitivity to the number of curriculum updates.

Number of updates	2	3	4	5
Final score	0.816	0.813	0.818	0.825

C.4. Ablation Study on Quick-answer Prompting

As shown in Sec. 5.4, our prefix-token entropy estimation correlates well with full-sequence estimation. This accuracy is critically supported by the use of Quick-answer Prompting (QAP). We conducted an ablation study to demonstrate its importance.

The purpose of QAP is to push the model to begin generating the substantive part of its answer within the prefix window (e.g., the first 128 tokens). Without QAP, the model might use the entire prefix to simply rephrase the question or begin a lengthy preamble, delaying the actual answer. In such cases, the prefix entropy would only reflect the model’s uncertainty about the question’s phrasing, not its uncertainty about the underlying answer, making it a poor proxy for sample difficulty.

As shown in Fig. 5, when QAP is removed, the rank-alignment between prefix-based and full-sequence entropy degrades substantially. This confirms that QAP is essential for making the prefix-token entropy a reliable and effective signal for our dynamic curriculum.

C.5. Curriculum Difficulty from Different CL Methods

To better understand the behavior of different CL strategies, we analyzed the intrinsic difficulty of the samples selected by each method at the beginning of training. We measured difficulty by evaluating the Qwen3-1.7B model’s accuracy on the first batch of samples chosen by each curriculum.

As shown in Table 13, EDCO and AC select the most difficult samples, with the base model achieving low accuracy on them. However, the extremely low accuracy from AC’s samples demonstrates that answer length fails to serve as the signal to indicate the problem difficulty in this setting. In contrast, length-based (Length) and perplexity-based (PPL) curricula select comparatively easier samples.

C.6. Curriculum Orchestration with Moderate-entropy Window

To investigate whether selecting the samples with highest entropy arise from nonsensical edge cases or OOD errors, which is harmful to model optimization, we conduct a “Moderate-entropy” experiment. Instead of selecting the top- N highest entropy samples (Top 0-6.67%), we selected a “Moderate” window (e.g., Top 5-11.67%) and compared the fine-tuning performance. We conducted experiments on Datacom domain with Qwen3-4B, as the results shown in the Tab. 14.

The original EDCO outperform the Moderate-entropy strategy by 2.18%. We have two main conclusions from this experiment: (1) EDCO does not suffer much from the high entropy caused by nonsensical edge cases or OOD errors. One potential reason is that the dataset used has been preprocessed to filter the sample with clear writing issues; (2) selecting samples with highest inference entropy is a valid approach to LLM fine-tuning, based on the results of selecting with moderate entropy window.

Table 12. Sensitivity to prefix length for entropy estimation.

Prefix length	No Training	20	50	100	128
Score	34.35	43.91	44.78	44.78	46.96

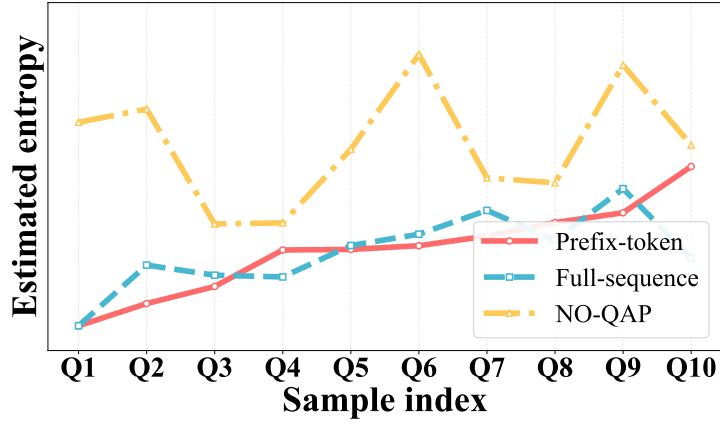


Figure 5. Ablation study on the quick-answer prompting. Prefix-token entropy estimation has a strong correlation with the full-length estimation (see blue and red lines in the figure).

Table 13. Base model accuracy on the initial training batch selected by different curriculum strategies. Lower accuracy indicates a selection of harder, more informative samples for the pre-trained model.

Method	Accuracy
EDCO	20%
Length	38.75%
AC	3.75%
PPL	31.25%

Table 14. Experiments with moderate-entropy for sample selection.

	No training	RS	Top 5-11.67%	Top 0-6.67%
Score	40	40.43	44.78	46.96